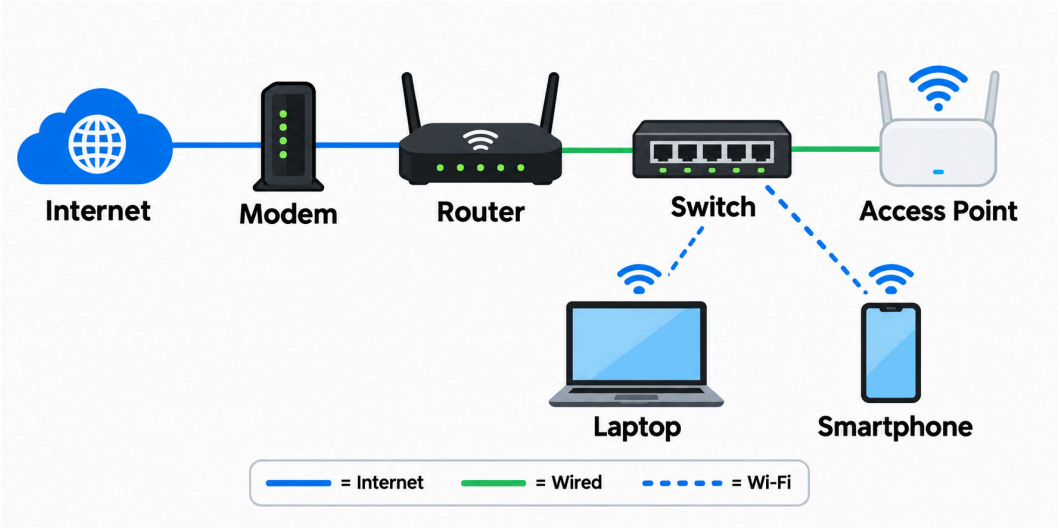
Assignment-4.1

“Networking hardware”

TASK-1 Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



TASK-2 List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

1. Modem (Modulator-Demodulator)

- **Main Function:** A modem connects your home or office to the internet service provided by your ISP. It converts signals so that data can travel between your internet connection and your devices.
- **Real-Life Example:** A modem connects to the internet line coming into your home and receives the internet connection from your service provider.

2. Router

- **Main Function:** A router connects your home network to the internet and shares the connection with multiple devices. It sends data to the correct destination.
- **Real-Life Example:** A home router allows your laptop, smartphone, and smart TV to use the same internet connection.

3. Switch (Network Switch)

- **Main Function:** A switch connects multiple wired devices within the same network. It sends data to the correct device using its MAC address.

- **Real-Life Example:** In an office, a switch connects several computers so they can share files and communicate with each other.

4. Access Point (Wireless Access Point)

- **Main Function:** An access point provides Wi-Fi connectivity to wireless devices and connects them to a wired network.
- **Real-Life Example:** A college installs access points in classrooms and libraries to provide Wi-Fi access to students and staff.

5. Network Cable

- **Main Function:** A network cable connects devices and carries data between them. Ethernet cables are commonly used for wired network connections.
- **Real-Life Example:** An Ethernet cable connects a desktop computer to a router, providing a stable internet connection.

TASK-3 Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

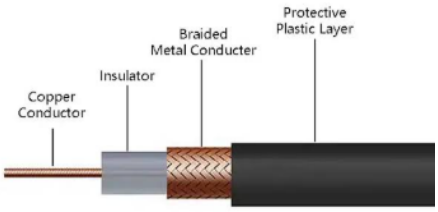
1. Ethernet Cable (Twisted-Pair Cable)

- **Use:** Ethernet cables are commonly used to connect computers, routers, and network switches in homes, schools, and offices. They provide a reliable connection for sharing data and accessing the internet.



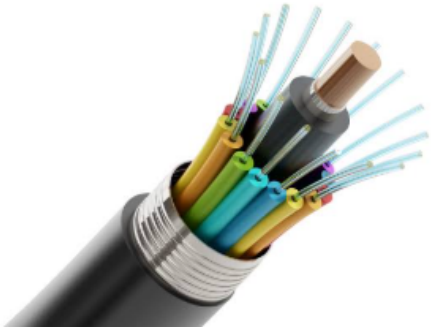
2. Coaxial Cable

- Use: Coaxial cables are commonly used for cable television, CCTV camera systems, and cable broadband internet connections. They carry signals between devices and help provide a stable connection.



3. Fiber-Optic Cable

- Use: Fiber-optic cables are used for high-speed internet, telecommunications, and sending large amounts of data over long distances. They use light signals to transfer data quickly.



TASK-4 Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature / Aspect	Router	Switch
1. Main Use	Connects multiple distinct networks together (such as a home or office network to the wider Internet) and directs traffic between them.	Connects multiple devices (like computers, printers, and gaming PCs) together inside a single local area network (LAN).
2. Typical Location	Positioned at the edge of the network where the local network meets the external ISP (Internet Service Provider) line.	Positioned inside the local network infrastructure to expand the number of available wired ports for local devices.
3. How They Handle Data (Addressing)	Works at Layer 3 (Network Layer) using **IP addresses** to forward data packets across different networks and choose the best route.	Works at Layer 2 (Data Link Layer) using **MAC addresses** to forward data frames directly to the specific destination port within the same local network.
4. Traffic Control & Scope	Acts as a traffic gateway/policer for external communication, handling tasks like Network Address Translation (NAT) and firewall filtering.	Acts as an internal high-speed multi-way switchboard, allowing local devices to communicate simultaneously without slowing down the router.